🎓 AWESOME CYBER SECURITY HEAVEN · REALM DOSSIER · 2026 EDITION

The Academy

Cybersecurity Learning Resources & Career Paths

24	165,193	42%	7
FREE TOOLS	COMBINED STARS	ON GITHUB	ACTIVE IN 2026

Intro: Learn the craft — free academies, knowledge bases, walkthroughs and career roadmaps.

Deep-research edition. Every entry re-verified and enriched with live GitHub statistics (stars, first release, last push, license, language) pulled 2026-10-17 via the GitHub API, plus per-realm field intelligence. Tool profiles rank the realm's most-adopted projects and note who uses each tool and the commercial product it replaces; the full inventory keeps every curated entry.

Part of Awesome Cyber Security Heaven — **444 curated free tools across 14 realms**.

github.com/RanaKhan77/awesome-cyber-security-heaven
CC0 · 2026-10-17

01 Realm Intelligence

live GitHub census · 2026-10-17

24	10	165,193	7	9,763	2014
CURATED TOOLS	ON GITHUB	COMBINED STARS	PUSHED 2026	MEDIAN STARS	OLDEST ENTRY

REALM BRIEF

Intro: Learn the craft — free academies, knowledge bases, walkthroughs and career roadmaps.

FIELD INTELLIGENCE (AI-ASSISTED)

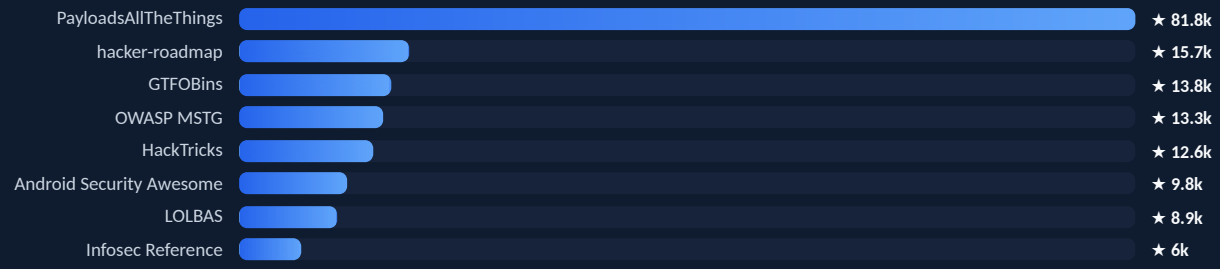
The Academy realm of cybersecurity learning resources and career paths curates 24 tools, with 10 hosted on GitHub representing 42% of the total collection. Adoption is heavily concentrated in top-tier tools, as the combined star count of all curated tools reaches 165,193, with a median star rating of 9,763, and the five highest-starred entries accounting for a large share of this total, led by PayloadsAllTheThings at 81.8k stars. Maintenance activity shows limited recent engagement, with only 7 of the 10 GitHub-hosted tools receiving code pushes in 2026, leaving three quiet classics—hacker-roadmap, Infosec Reference, and SecuriBench Micro—without updates that year. The toolset draws from a narrow range of programming languages, with Python being the most common, and uses a mix of licenses including MIT and Apache-2.0, while the oldest entry dates to July 22, 2014, indicating a long-running, established collection with uneven modern maintenance.

Synthesized by seed-2-0-mini over the live census: stars, first-release years, push activity and project claims pulled from the GitHub API on 2026-10-17. Human-curated realm, AI-summarized numbers.

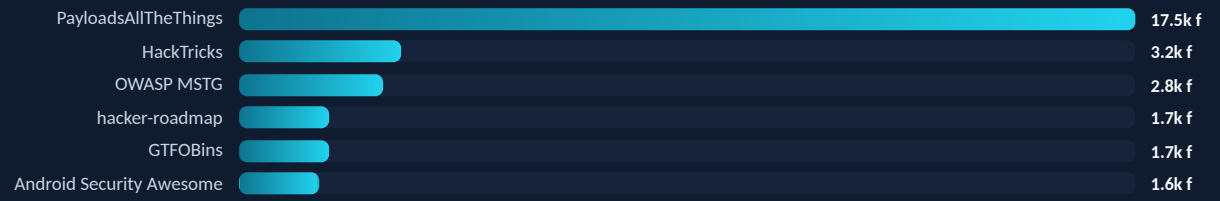
01 The Adoption Map

who the community actually stars

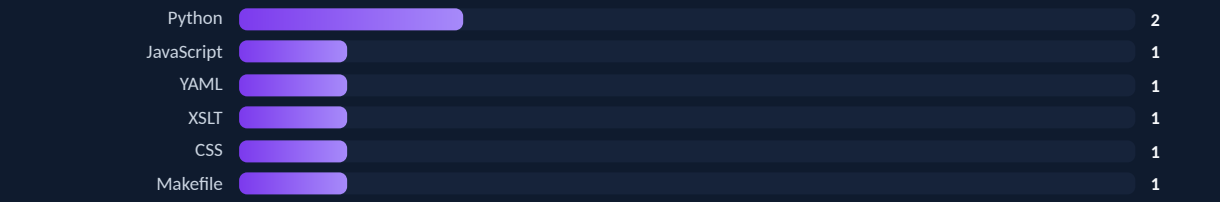
TOP TOOLS BY GITHUB STARS



MOST-FORKED — REAL-WORLD REUSE SIGNAL



LANGUAGE MIX — GITHUB-HOSTED ENTRIES



01 Generations & Momentum

the realm's age structure

THE FOUNDERS — OLDEST TRACKED ENTRIES

Android Security Awesome	2014-07-22
Infosec Reference	2014-09-13
SecuriBench Micro	2014-12-01
OWASP MSTG	2016-09-30
PayloadsAllTheThings	2016-10-18

THE NEWCOMERS — NEWEST GITHUB ARRIVALS

Network Segmentation Cheat Sheet	2021-11
HackTricks	2020-07
LOLBAS	2018-06
GTFOBins	2018-05
hacker-roadmap	2018-04

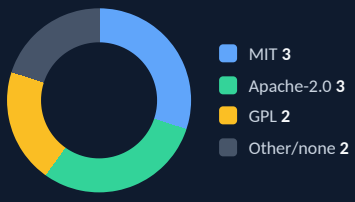
CREATION ERAS — WHEN THE REALM WAS BUILT

2008-2012	2013-2016	2017-2020	2021-2026
0	5	4	1

QUIET CLASSICS (NO 2026 PUSH, STILL THE STANDARD)

hacker-roadmap, Infosec Reference, SecuriBench Micro

LICENSE MIX — GITHUB-HOSTED ENTRIES



02 Tool Profiles — the heavyweights

full-page dossiers · ranked by adoption

Each heavyweight gets a full dossier: live GitHub census (stars, forks, first-release year, last push, license, language), the project's own claim, our curator's note, **who uses it and which commercial product it replaces**, and its star weight against the realm's #1.

Profiles are ordered by GitHub adoption (stars). Non-GitHub entries follow, alphabetically. Flagship marks the realm's top 8 by stars.

01 PayloadsAllTheThings

★ FLAGSHIP PICK ACTIVE 2026

github.com/swisskyrepo/PayloadsAllTheThings

star weight vs realm #1



★ 81.8k	GITHUB STARS
17,475	FORKS
2026-08	LAST PUSH
2016	FIRST DEVELOPED
MIT	LICENSE
Python	LANGUAGE

THEIR CLAIM — AS THE PROJECT STATES IT

"A list of useful payloads and bypass for Web Application Security and Pentest/CTF"

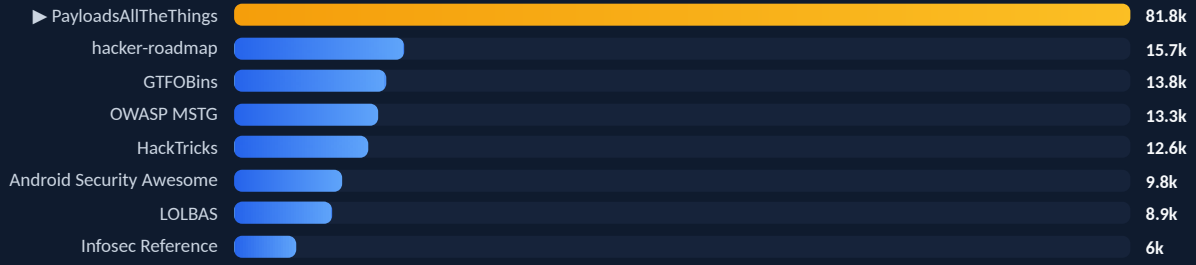
CURATOR'S NOTE — WHY IT EARNS A PLACE

the curated payload and methodology repo — copy, paste, understand, learn.

WHO USES IT · WHAT IT REPLACES

Used by pentesters grabbing working payloads fast; the community cookbook for every attack class.

WHERE IT SITS — REALM TOP 8 BY STARS (▶ = THIS TOOL)



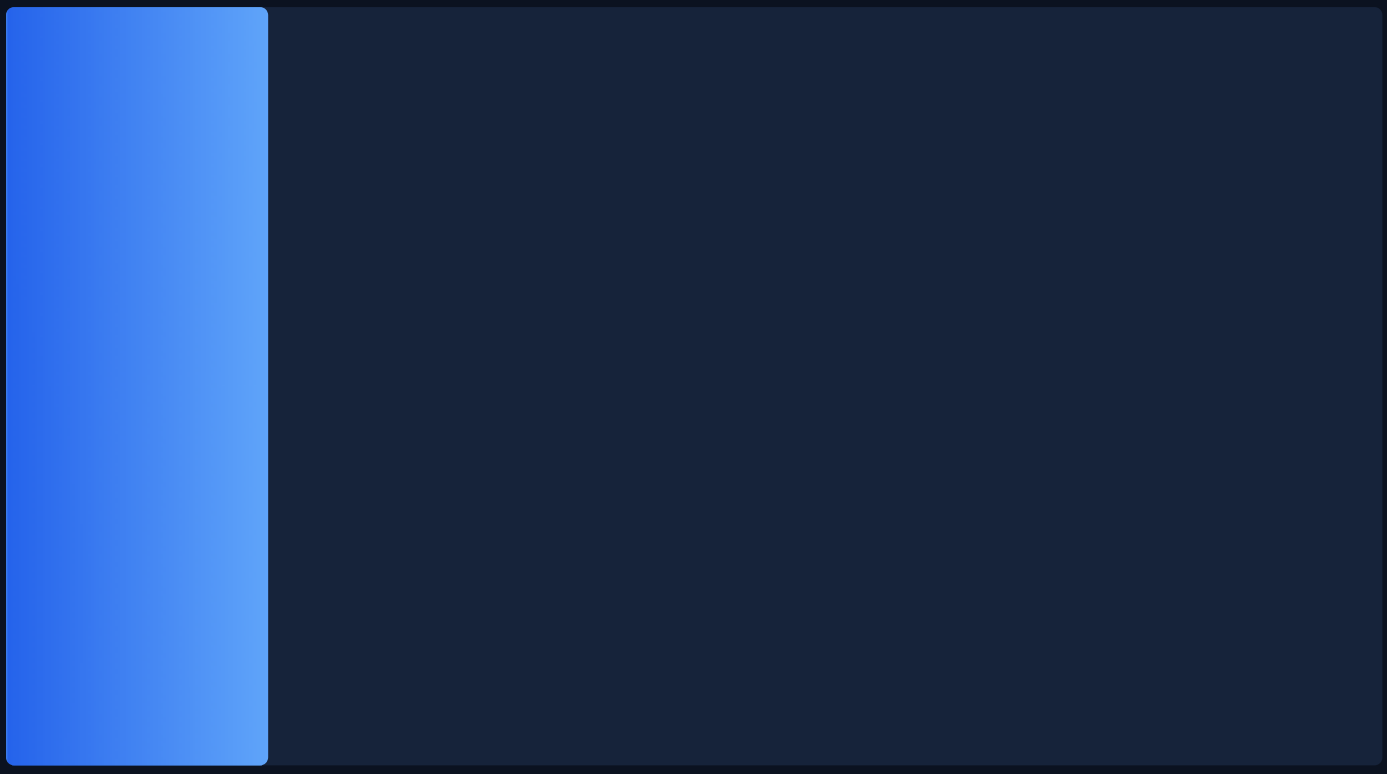
10	17,475	38	yes
YEARS ON GITHUB	FORKS (REUSE)	OPEN ISSUES	2026 MOMENTUM

02 hacker-roadmap

github.com/sundowndev/hacker-roadmap

★ FLAGSHIP PICK

star weight vs realm #1



★ 15.7k	GITHUB STARS
1,717	FORKS
2023-10	LAST PUSH
2018	FIRST DEVELOPED
MIT	LICENSE
—	LANGUAGE

THEIR CLAIM — AS THE PROJECT STATES IT

"A collection of hacking tools, resources and references to practice ethical hacking."

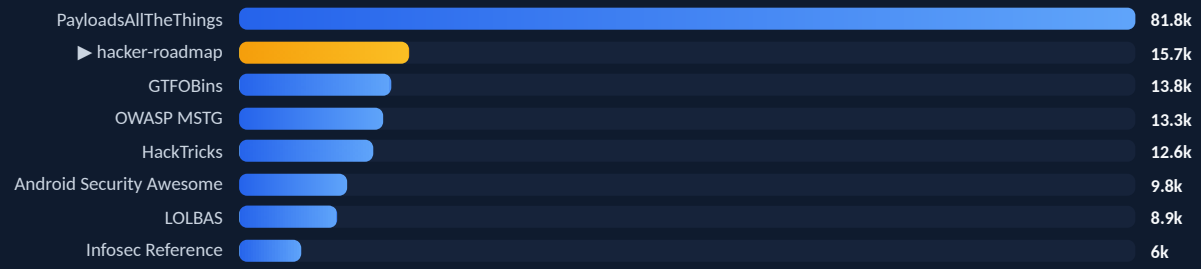
CURATOR'S NOTE — WHY IT EARNS A PLACE

the illustrated cybersecurity study/career roadmap; the "where do I start?" answer.

WHO USES IT · WHAT IT REPLACES

Used by beginners planning their path into security; the free curated curriculum for self-learners.

WHERE IT SITS — REALM TOP 8 BY STARS (▶ = THIS TOOL)



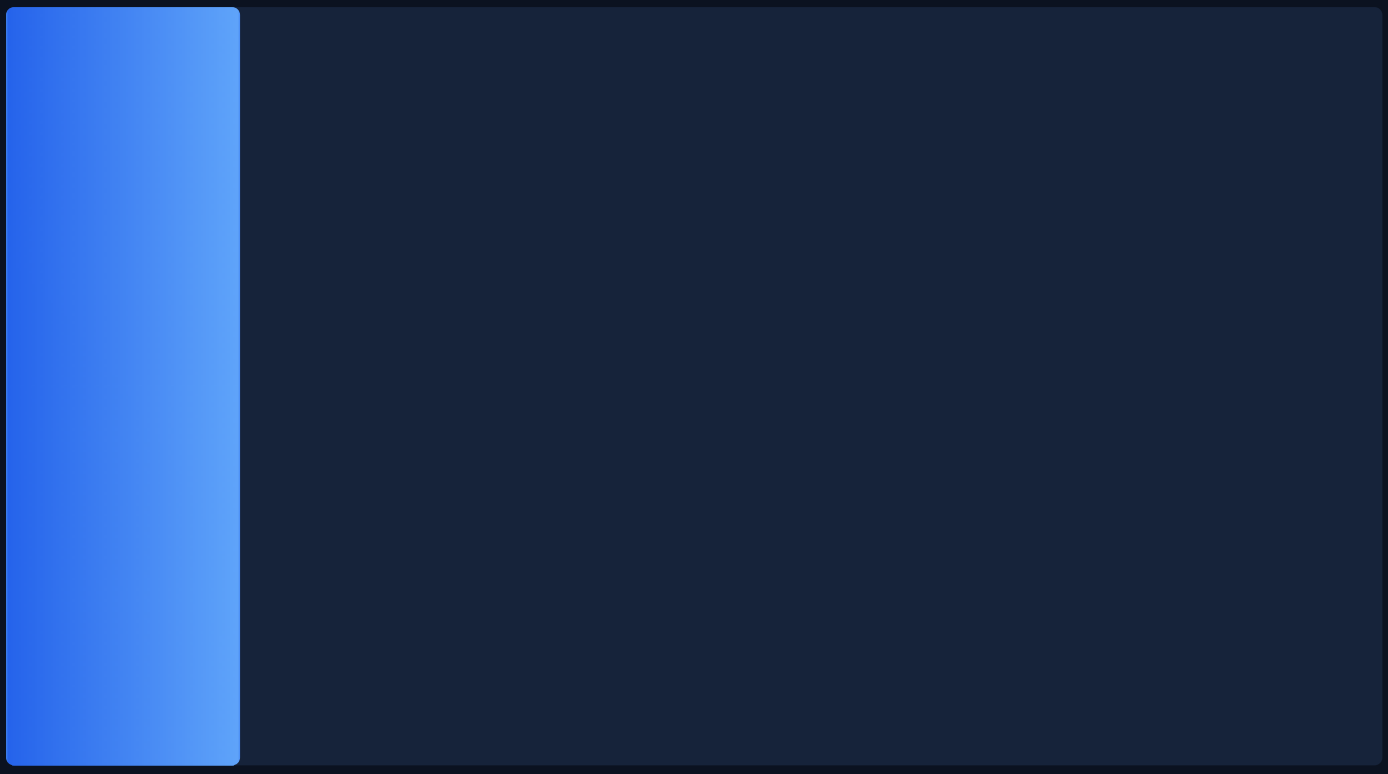
8	1,717	3	—
YEARS ON GITHUB	FORKS (REUSE)	OPEN ISSUES	2026 MOMENTUM

03 GTFOBins

github.com/GTFOBins/GTFOBins.github.io

★ FLAGSHIP PICK
ACTIVE 2026

star weight vs realm #1



★ 13.8k	GITHUB STARS
1,705	FORKS
2026-05	LAST PUSH
2018	FIRST DEVELOPED
GPL-3.0	LICENSE
YAML	LANGUAGE

THEIR CLAIM — AS THE PROJECT STATES IT

"GTFOBins is a curated list of Unix-like executables that can be used to bypass local security restrictions in misconfigured systems."

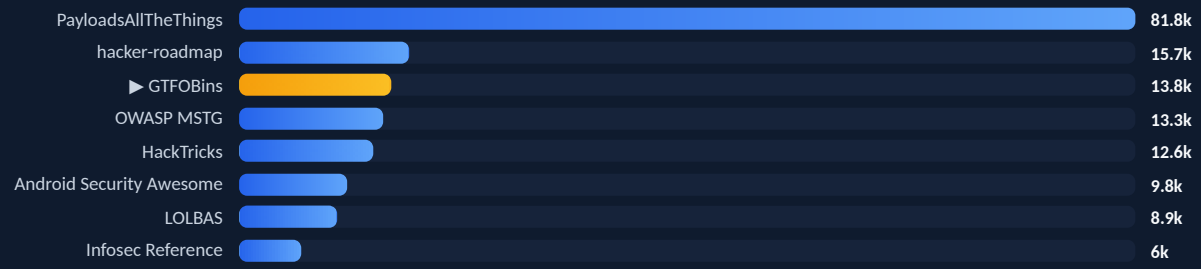
CURATOR'S NOTE — WHY IT EARNS A PLACE

the living Linux binary-abuse matrix; the privilege-escalation bible.

WHO USES IT · WHAT IT REPLACES

Used by pentesters finding Unix binary-abuse paths for privesc; the standard free lookup red teams quote.

WHERE IT SITS — REALM TOP 8 BY STARS (▶ = THIS TOOL)



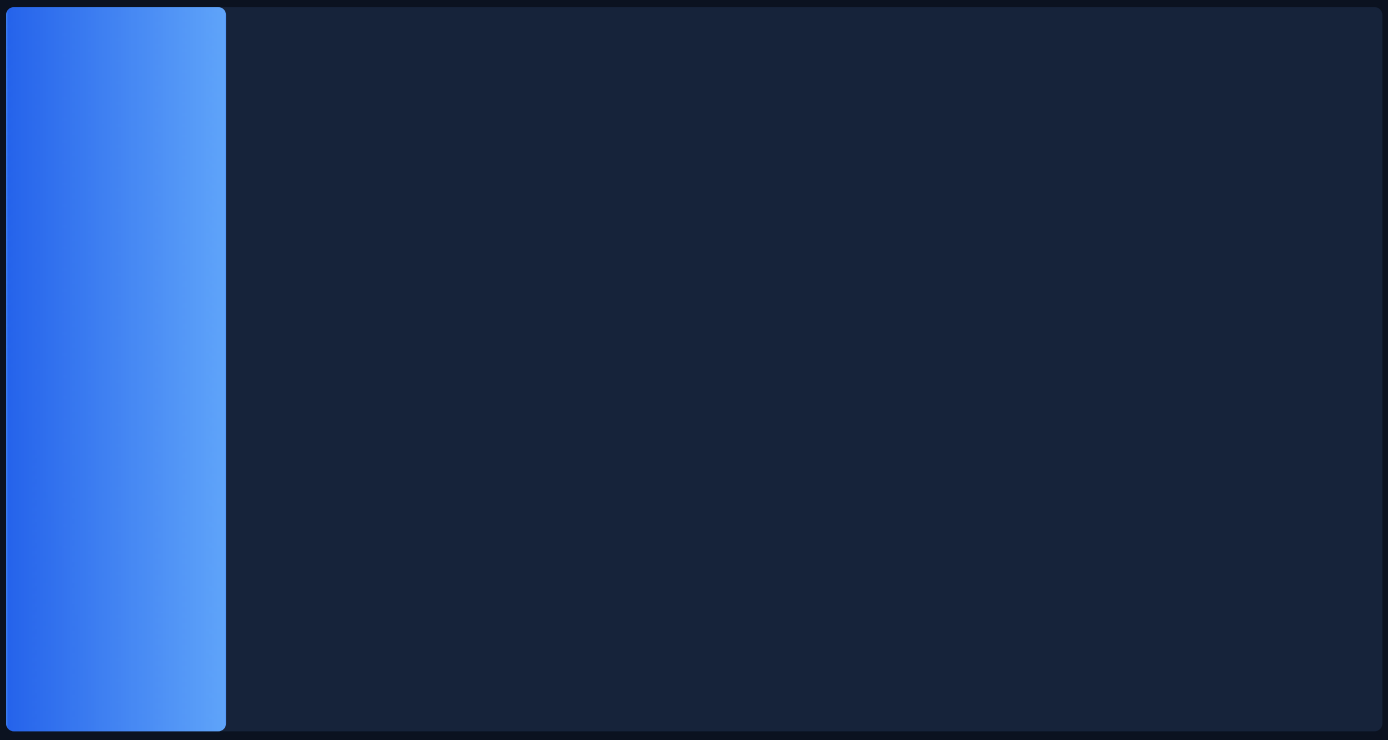
8	1,705	31	yes
YEARS ON GITHUB	FORKS (REUSE)	OPEN ISSUES	2026 MOMENTUM

04 OWASP MSTG

github.com/OWASP/owasp-mstg

★ FLAGSHIP PICK ACTIVE 2026

star weight vs realm #1



★ 13.3k	GITHUB STARS
2,828	FORKS
2026-10	LAST PUSH
2016	FIRST DEVELOPED
CC-BY-SA-4.0	LICENSE
Python	LANGUAGE

THEIR CLAIM — AS THE PROJECT STATES IT

"The OWASP Mobile Application Security Testing Guide (MASTG) is a comprehensive manual for mobile app security testing and reverse engineering. It describes technical processes for verifying the OWASP Mobile Security Weak"

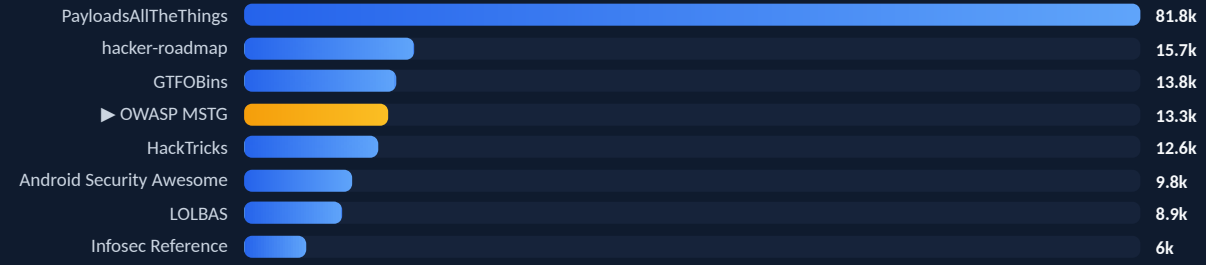
CURATOR'S NOTE — WHY IT EARNS A PLACE

the Mobile Security Testing Guide: the canonical methodology for mobile app assessments.

WHO USES IT · WHAT IT REPLACES

Used by mobile security testers following the standard methodology; the OWASP mobile testing guide commercial mobile audits follow.

WHERE IT SITS — REALM TOP 8 BY STARS (▶ = THIS TOOL)



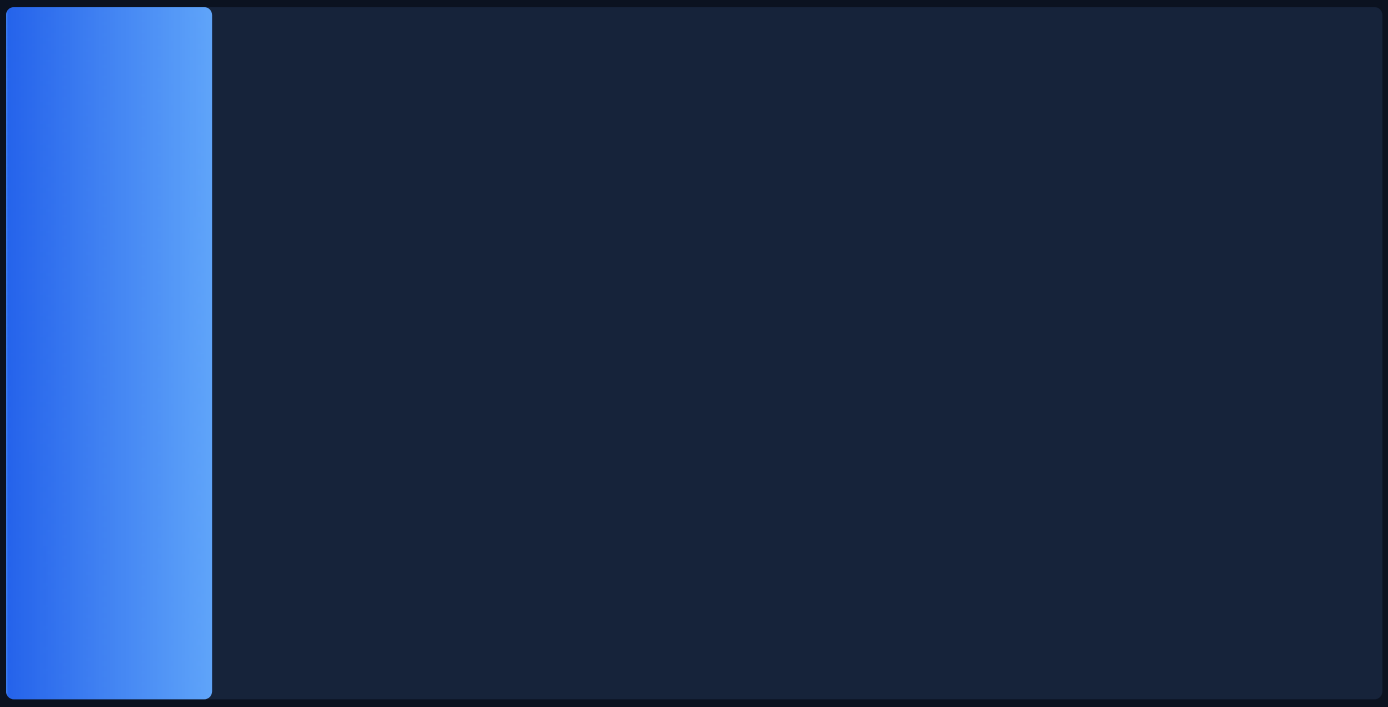
10	2,828	245	yes
YEARS ON GITHUB	FORKS (REUSE)	OPEN ISSUES	2026 MOMENTUM

05 HackTricks

github.com/HackTricks-wiki/hacktricks

★ FLAGSHIP PICK ACTIVE 2026

star weight vs realm #1



★ 12.6k	GITHUB STARS
3,203	FORKS
2026-10	LAST PUSH
2020	FIRST DEVELOPED
None	LICENSE
JavaScript	LANGUAGE

THEIR CLAIM — AS THE PROJECT STATES IT

"Welcome to the page where you will find each trick/technique/whatever I have learnt in CTFs, real life apps, and reading researches and news."

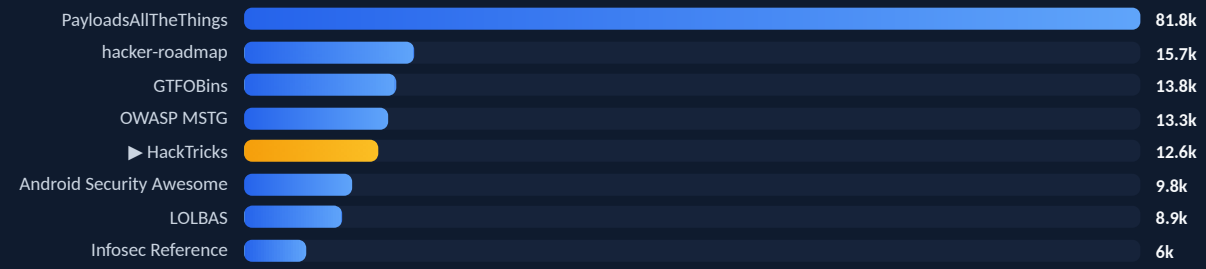
CURATOR'S NOTE — WHY IT EARNS A PLACE

the pentester's playbook: technique-by-technique attack references from recon to post-exploitation.

WHO USES IT · WHAT IT REPLACES

Used by pentesters as the encyclopedic reference during engagements; the donation-supported playbook the community keeps current. (free; custom content license)

WHERE IT SITS — REALM TOP 8 BY STARS (▶ = THIS TOOL)



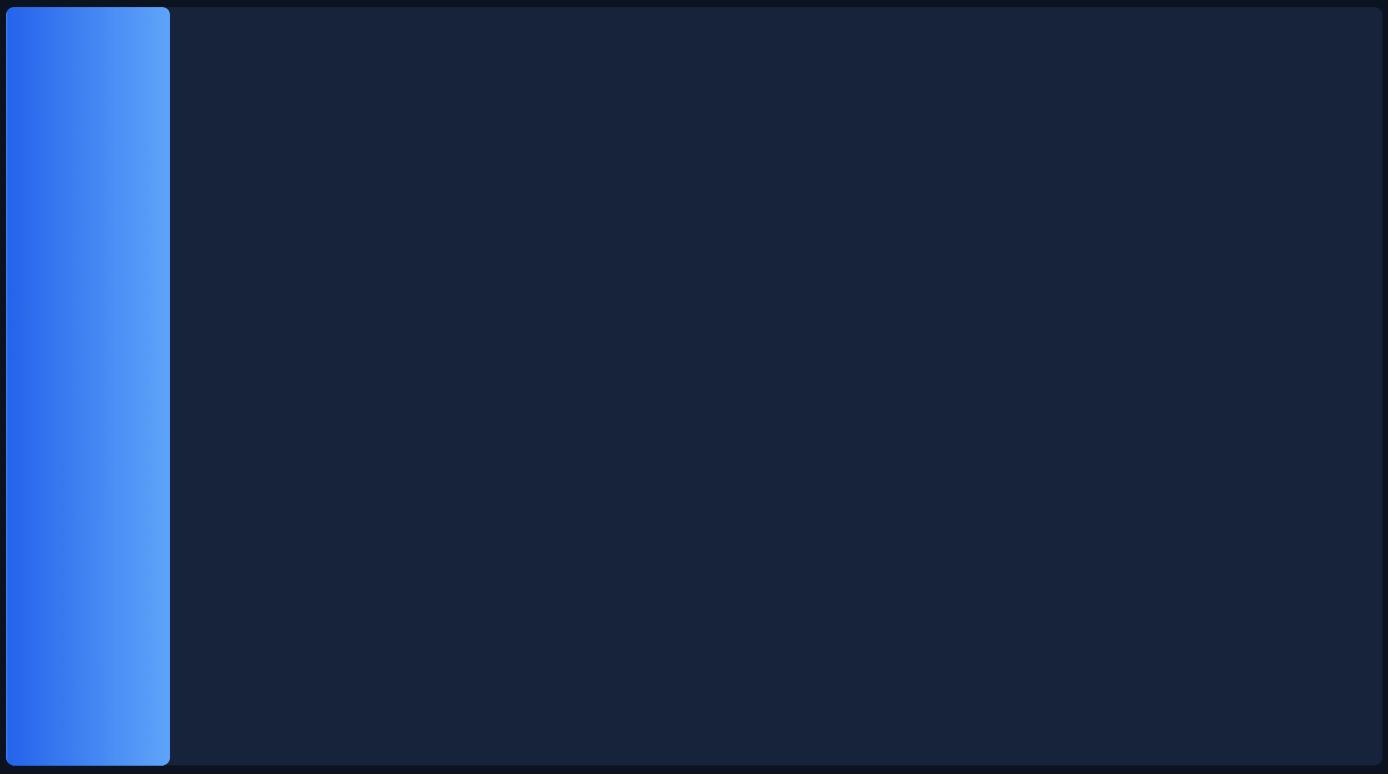
6	3,203	2	yes
YEARS ON GITHUB	FORKS (REUSE)	OPEN ISSUES	2026 MOMENTUM

06 Android Security Awesome

★ FLAGSHIP PICK ACTIVE 2026

github.com/ashishb/android-security-awesome

star weight vs realm #1



★ 9.8k	GITHUB STARS
1,584	FORKS
2026-10	LAST PUSH
2014	FIRST DEVELOPED
Apache-2.0	LICENSE
Makefile	LANGUAGE

THEIR CLAIM — AS THE PROJECT STATES IT

"A collection of android security related resources"

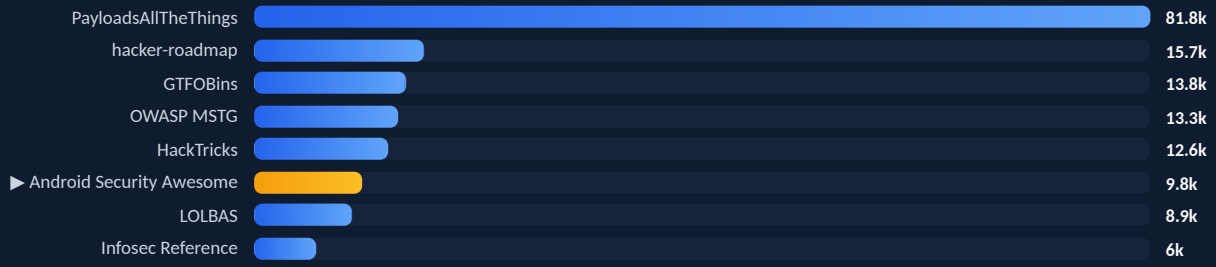
CURATOR'S NOTE — WHY IT EARNS A PLACE

curated Android security research: tools, papers and training.

WHO USES IT · WHAT IT REPLACES

Used by mobile security researchers assembling toolkits.

WHERE IT SITS — REALM TOP 8 BY STARS (▶ = THIS TOOL)



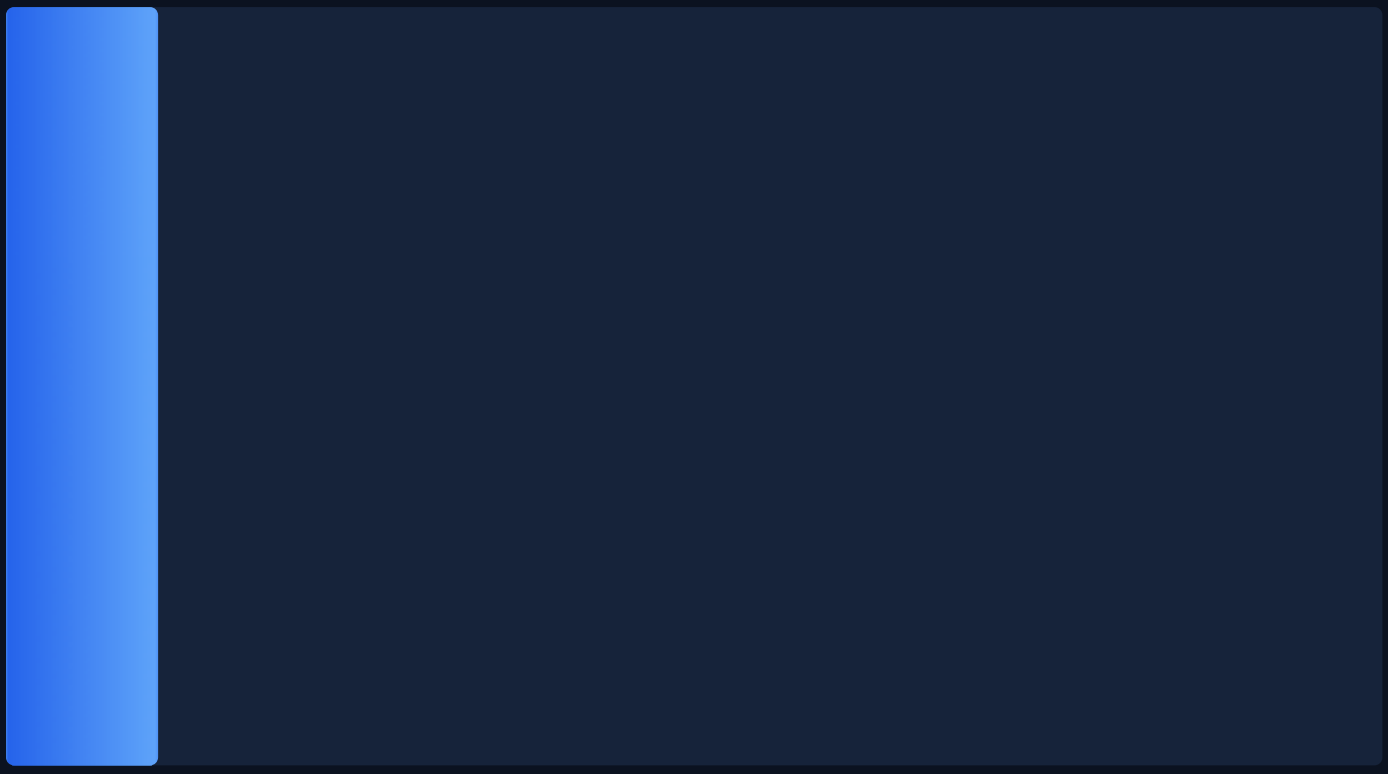
12	1,584	0	yes
YEARS ON GITHUB	FORKS (REUSE)	OPEN ISSUES	2026 MOMENTUM

07 LOLBAS

github.com/LOLBAS-Project/LOLBAS

★ FLAGSHIP PICK ACTIVE 2026

star weight vs realm #1



★ 8.9k	GITHUB STARS
1,182	FORKS
2026-09	LAST PUSH
2018	FIRST DEVELOPED
GPL-3.0	LICENSE
XSLT	LANGUAGE

THEIR CLAIM — AS THE PROJECT STATES IT

"Living Off The Land Binaries And Scripts - (LOLBins and LOLScripts)"

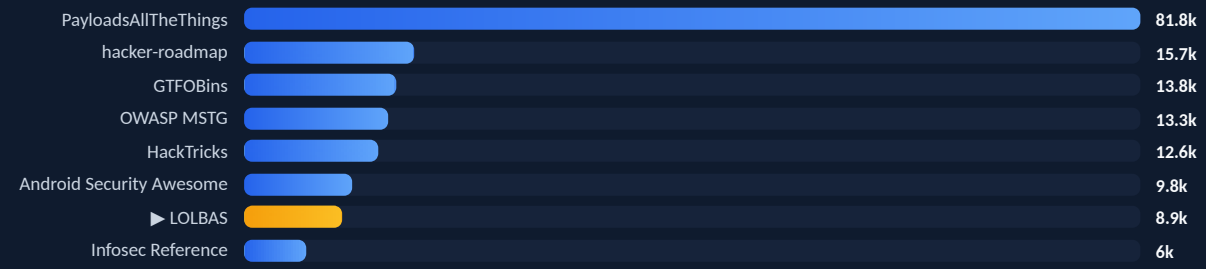
CURATOR'S NOTE — WHY IT EARNS A PLACE

the Windows counterpart to GTFOBins: living-off-the-land binaries and scripts.

WHO USES IT · WHAT IT REPLACES

Used by red teamers planning fileless attacks and defenders writing detections for them.

WHERE IT SITS — REALM TOP 8 BY STARS (▶ = THIS TOOL)



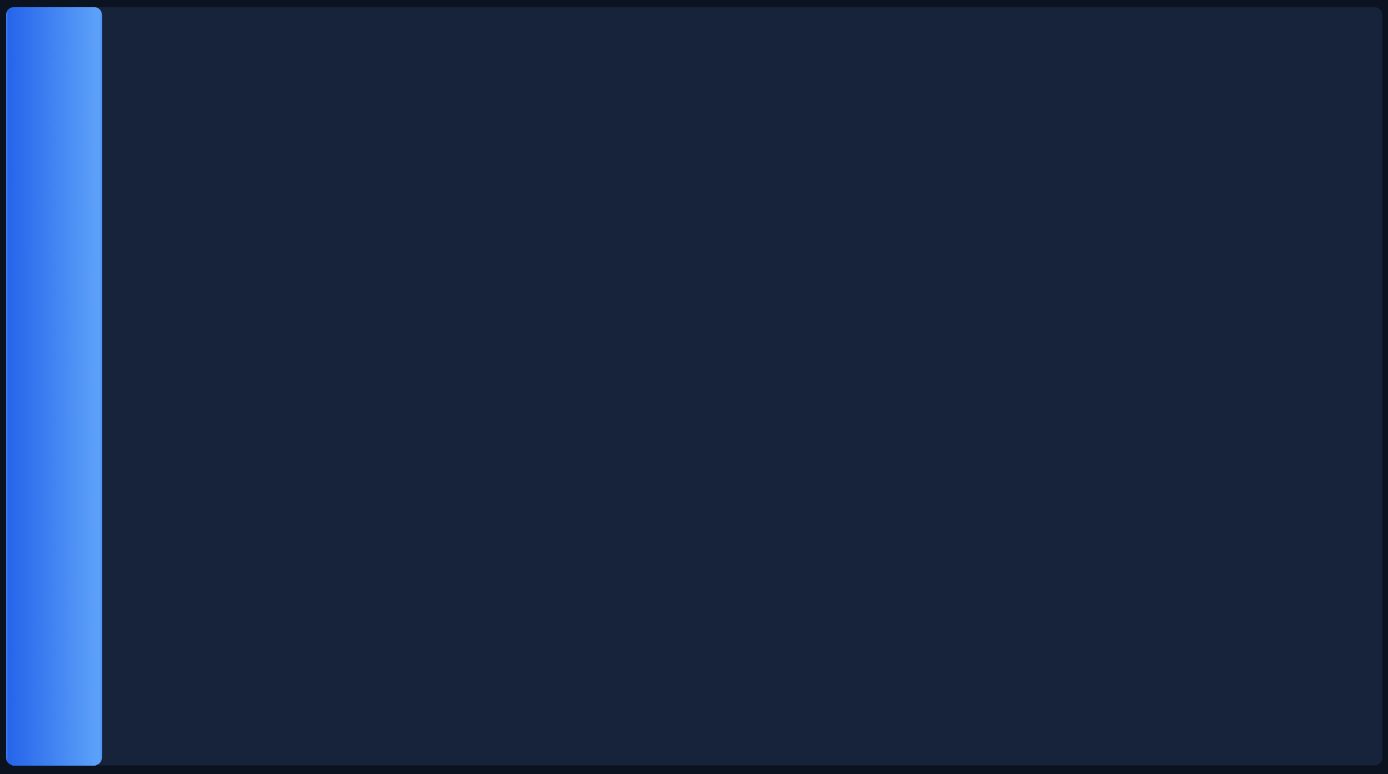
8	1,182	29	yes
YEARS ON GITHUB	FORKS (REUSE)	OPEN ISSUES	2026 MOMENTUM

08 Infosec Reference

github.com/rmusser01/Infosec_Reference

★ FLAGSHIP PICK

star weight vs realm #1



★ 6k	GITHUB STARS
1,230	FORKS
2025-10	LAST PUSH
2014	FIRST DEVELOPED
MIT	LICENSE
CSS	LANGUAGE

THEIR CLAIM — AS THE PROJECT STATES IT

"An Information Security Reference That Doesn't Suck; https://rmusser.net/git/admin-2/Infosec_Reference for non-MS Git hosted version."

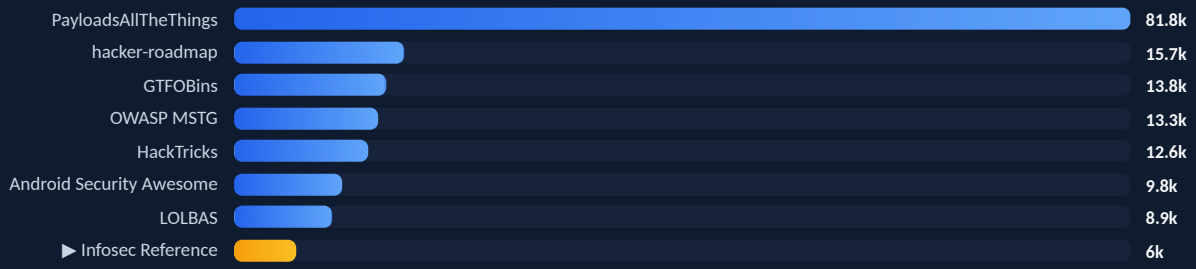
CURATOR'S NOTE — WHY IT EARNS A PLACE

the giant information-security reference: tools, papers and guides in one place.

WHO USES IT · WHAT IT REPLACES

Used by learners and practitioners organizing knowledge; vendor-marketing-free. (curated list)

WHERE IT SITS — REALM TOP 8 BY STARS (▶ = THIS TOOL)



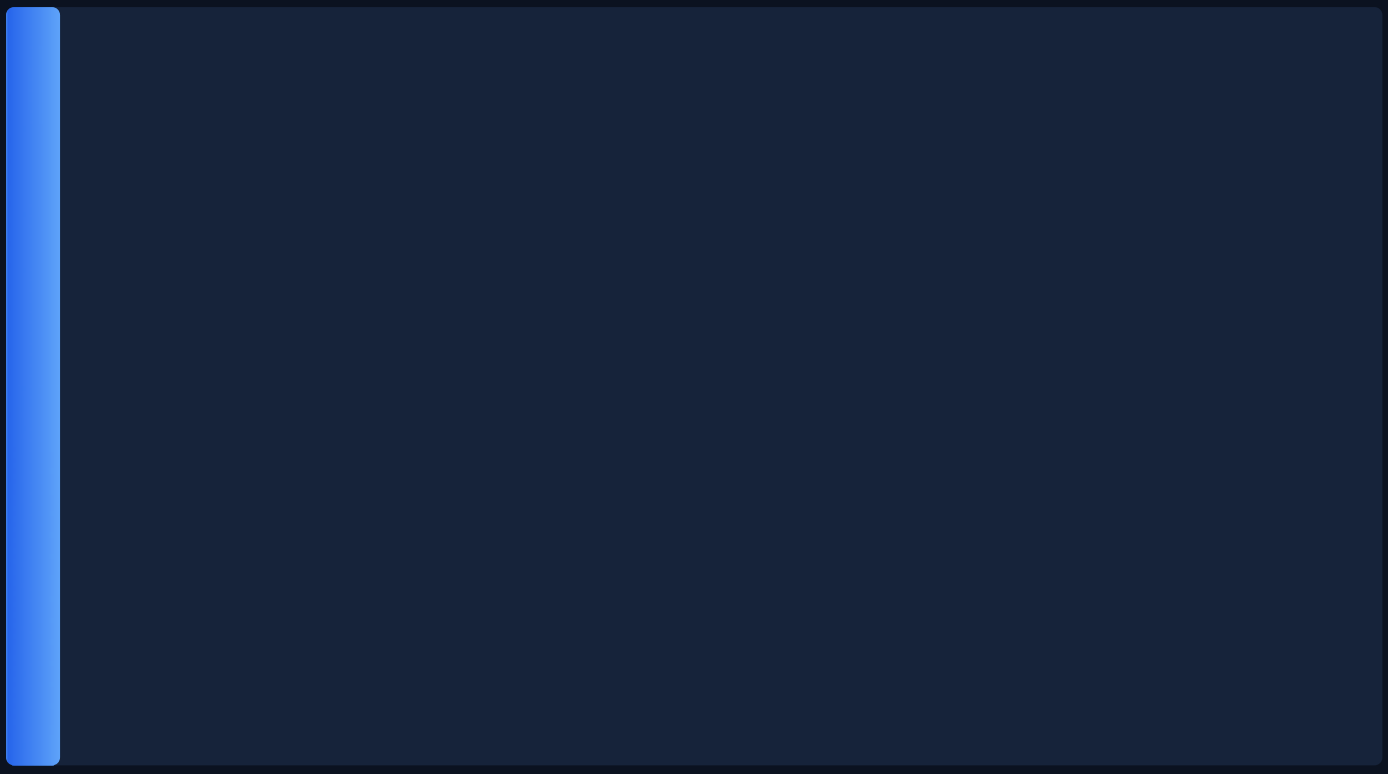
12	1,230	4	—
YEARS ON GITHUB	FORKS (REUSE)	OPEN ISSUES	2026 MOMENTUM

09 Network Segmentation Cheat Sheet

ACTIVE 2026

github.com/sergiomarotco/Network-segmentation-cheat-sheet

star weight vs realm #1



★ 3.5k	GITHUB STARS
286	FORKS
2026-02	LAST PUSH
2021	FIRST DEVELOPED
Apache-2.0	LICENSE
—	LANGUAGE

THEIR CLAIM — AS THE PROJECT STATES IT

"Best practices for segmentation of the corporate network of any company"

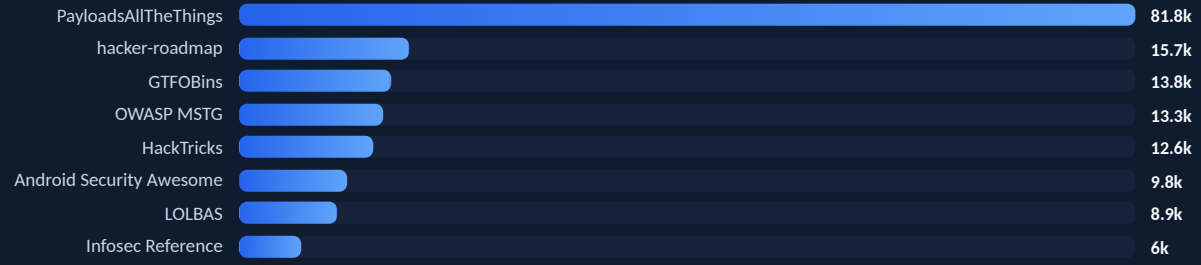
CURATOR'S NOTE — WHY IT EARNS A PLACE

practical segmentation architecture: VLANs, zones and rules that survive audits.

WHO USES IT · WHAT IT REPLACES

Used by network architects designing defensible segmentation; the free practical reference.

WHERE IT SITS — REALM TOP 8 BY STARS (▶ = THIS TOOL)

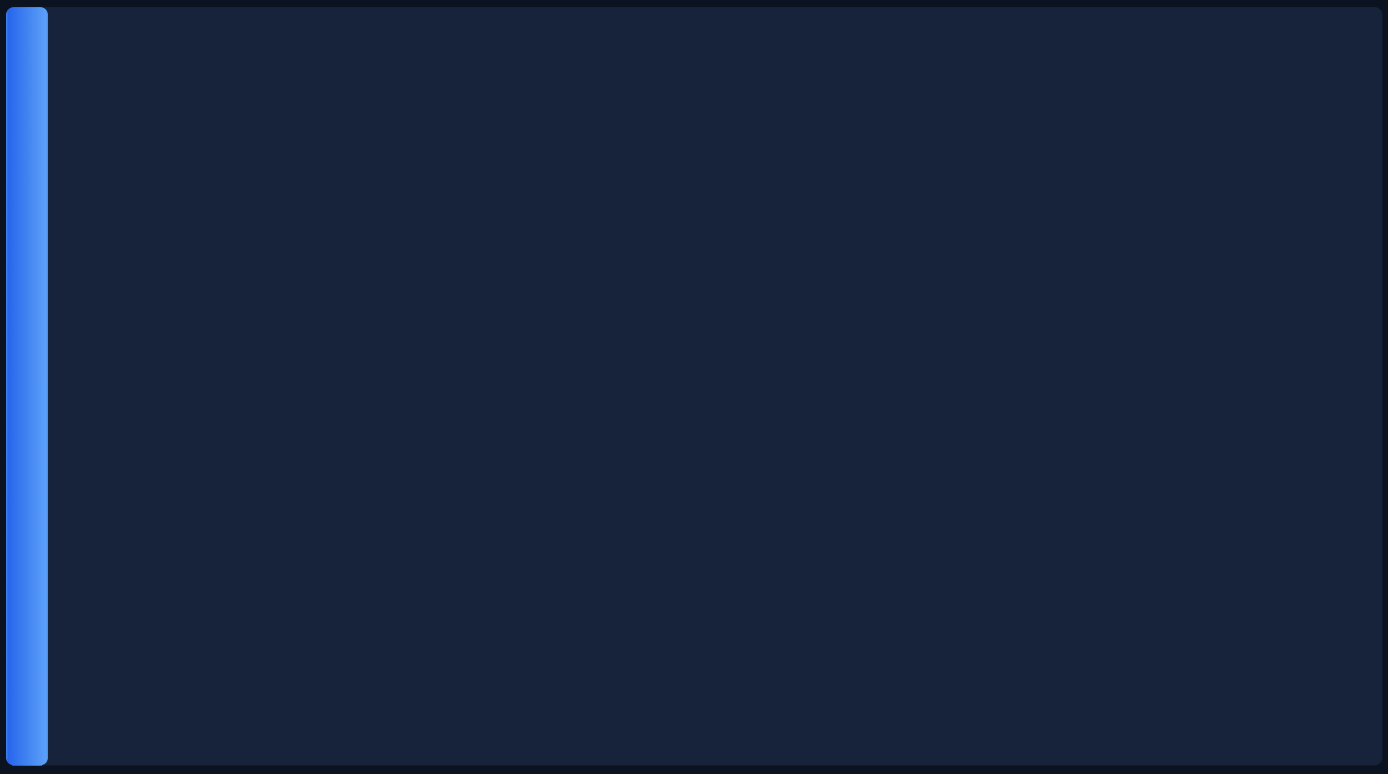


5	286	2	yes
YEARS ON GITHUB	FORKS (REUSE)	OPEN ISSUES	2026 MOMENTUM

10 SecuriBench Micro

github.com/too4words/securibench-micro/

star weight vs realm #1



★ 27	GITHUB STARS
24	FORKS
2018-07	LAST PUSH
2014	FIRST DEVELOPED
Apache-2.0	LICENSE
Java	LANGUAGE

THEIR CLAIM — AS THE PROJECT STATES IT

"Securibench Micro is a benchmark for static analysis tools for security."

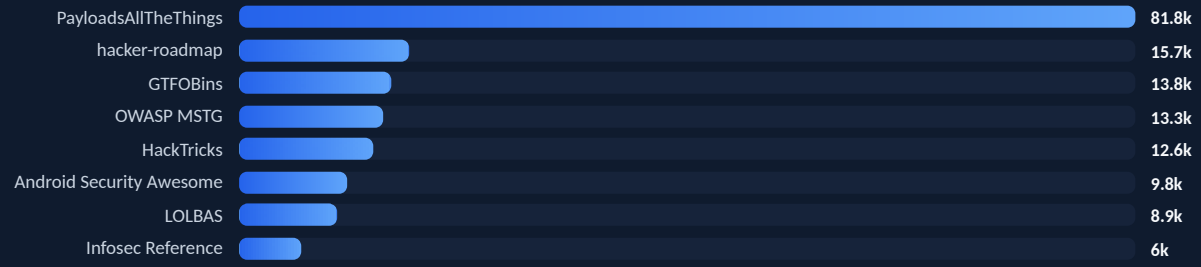
CURATOR'S NOTE — WHY IT EARNS A PLACE

the Stanford benchmark suite for security-analysis tools.

WHO USES IT · WHAT IT REPLACES

Used by academic researchers benchmarking static analysis; Stanford's free test suite for tool evaluation.

WHERE IT SITS — REALM TOP 8 BY STARS (▶ = THIS TOOL)



12	24	5	—
YEARS ON GITHUB	FORKS (REUSE)	OPEN ISSUES	2026 MOMENTUM

11 Bug Bounty Hunter

www.bugbountyhunter.com/

WEB / STANDALONE

NOT GITHUB-HOSTED · WEBSITE / STANDALONE

THEIR CLAIM — AS THE PROJECT STATES IT

"Bug Bounty Hunter"

CURATOR'S NOTE — WHY IT EARNS A PLACE

free bug-bounty training platform: real methodology and lab targets.

WHO USES IT · WHAT IT REPLACES

Used by aspiring hunters learning methodology; the free resource for bounty skill-building. (free service)

12 CTF Field Guide

trailofbits.github.io/ctf/

WEB / STANDALONE

—

NOT GITHUB-HOSTED · WEBSITE / STANDALONE

THEIR CLAIM — AS THE PROJECT STATES IT

"CTF Field Guide"

CURATOR'S NOTE — WHY IT EARNS A PLACE

Trail of Bits' field guide to capture-the-flag: from fundamentals to finals.

WHO USES IT · WHAT IT REPLACES

Used by CTF newcomers learning the game; the free guide from the CTF community.

13 Cyber Security Base

cybersecuritybase.github.io/

WEB / STANDALONE



—

NOT GITHUB-HOSTED · WEBSITE / STANDALONE

THEIR CLAIM — AS THE PROJECT STATES IT

"Cyber Security Base"

CURATOR'S NOTE — WHY IT EARNS A PLACE

free course series by Aalto University and F-Secure; from beginner to blue team.

WHO USES IT · WHAT IT REPLACES

Used by students taking university-level security courses; the free MOOC series from F-Secure and Helsinki. (free course)

14 Hacker101

www.hacker101.com/

WEB / STANDALONE

—

NOT GITHUB-HOSTED · WEBSITE / STANDALONE

THEIR CLAIM — AS THE PROJECT STATES IT

“Hacker101”

CURATOR'S NOTE — WHY IT EARNS A PLACE

HackerOne's free web-security course: video lessons plus an always-on CTF for bug-bounty practice.

WHO USES IT · WHAT IT REPLACES

Used by aspiring bounty hunters starting on HackerOne; the free training with bounties attached.

15 Hacksplaining

www.hacksplaining.com/

WEB / STANDALONE

—

NOT GITHUB-HOSTED · WEBSITE / STANDALONE

THEIR CLAIM — AS THE PROJECT STATES IT

"Hacksplaining"

CURATOR'S NOTE — WHY IT EARNS A PLACE

interactive security lessons for developers: exploit each vulnerability class, then patch it yourself.

WHO USES IT · WHAT IT REPLACES

Used by developers learning web vulns interactively; the free lesson series engineering teams assign.

16 IppSec

ippsec.rocks

WEB / STANDALONE

—

NOT GITHUB-HOSTED · WEBSITE / STANDALONE

THEIR CLAIM — AS THE PROJECT STATES IT

"IppSec"

CURATOR'S NOTE — WHY IT EARNS A PLACE

the legendary HTB walkthrough archive; watch a master work through every box, searchable by technique.

WHO USES IT · WHAT IT REPLACES

Used by HTB learners studying full walkthroughs; the free video walkthroughs that taught a generation of pentesters.

17 LiveOverflow

www.youtube.com/@LiveOverflow

WEB / STANDALONE

—

NOT GITHUB-HOSTED · WEBSITE / STANDALONE

THEIR CLAIM — AS THE PROJECT STATES IT

"LiveOverflow"

CURATOR'S NOTE — WHY IT EARNS A PLACE

deep-dive security videos that explain *why* things break — the channel every hacker watched.

WHO USES IT · WHAT IT REPLACES

Used by learners building deep intuition for exploitation; the free video series on hacking fundamentals.

18 OWASP Cheat Sheet Series

cheatsheetseries.owasp.org/

WEB / STANDALONE

—

NOT GITHUB-HOSTED · WEBSITE / STANDALONE

THEIR CLAIM — AS THE PROJECT STATES IT

"OWASP Cheat Sheet Series"

CURATOR'S NOTE — WHY IT EARNS A PLACE

concise, developer-friendly defensive guidance per topic; the fix side of every finding.

WHO USES IT · WHAT IT REPLACES

Used by developers building defenses correctly; the free OWASP guidance engineers link in code reviews.

19 OWASP WSTG

owasp.org/www-project-web-security-testing-guide/

WEB / STANDALONE

—

NOT GITHUB-HOSTED · WEBSITE / STANDALONE

THEIR CLAIM — AS THE PROJECT STATES IT

"OWASP WSTG"

CURATOR'S NOTE — WHY IT EARNS A PLACE

the Web Security Testing Guide — the methodology standard for thorough assessments.

WHO USES IT · WHAT IT REPLACES

Used by auditors and pentesters as the canonical assessment methodology; the free OWASP guide commercial audits map to.

20 PortSwigger Web Security Academy

portswigger.net/web-security

WEB / STANDALONE

—

NOT GITHUB-HOSTED · WEBSITE / STANDALONE

THEIR CLAIM — AS THE PROJECT STATES IT

"PortSwigger Web Security Academy"

CURATOR'S NOTE — WHY IT EARNS A PLACE

free, best-in-class web-hacking labs from the Burp Suite team; the standard web-security curriculum.

WHO USES IT · WHAT IT REPLACES

Used by aspiring bug bounty hunters and web pentesters; the free training platform from the Burp Suite makers that rivals paid courses.

21 Professor Messer

www.professormesser.com

WEB / STANDALONE

—

NOT GITHUB-HOSTED · WEBSITE / STANDALONE

THEIR CLAIM — AS THE PROJECT STATES IT

"Professor Messer"

CURATOR'S NOTE — WHY IT EARNS A PLACE

free complete video courses for Security+, Network+ and CySA+; the default free cert-prep resource.

WHO USES IT · WHAT IT REPLACES

Used by certification students as the default free alternative to paid bootcamps.

22 pwn.college

pwn.college

WEB / STANDALONE

—

NOT GITHUB-HOSTED · WEBSITE / STANDALONE

THEIR CLAIM — AS THE PROJECT STATES IT

"pwn.college"

CURATOR'S NOTE — WHY IT EARNS A PLACE

ASU's free hands-on curriculum from Linux basics to kernel exploitation, with dojo challenges.

WHO USES IT · WHAT IT REPLACES

Used by students mastering binary exploitation; Arizona State's free curriculum from basics to kernel pwn.

23 Roppers Academy

www.hoppersroppers.org/

WEB / STANDALONE

—

NOT GITHUB-HOSTED · WEBSITE / STANDALONE

THEIR CLAIM — AS THE PROJECT STATES IT

"Roppers Academy"

CURATOR'S NOTE — WHY IT EARNS A PLACE

free structured courses from computing fundamentals to security — the patient beginner's path.

WHO USES IT · WHAT IT REPLACES

Used by beginners building computing and security fundamentals; the free structured coursework.

24 SEED Labs

seedsecuritylabs.org/

WEB / STANDALONE

—

NOT GITHUB-HOSTED · WEBSITE / STANDALONE

THEIR CLAIM — AS THE PROJECT STATES IT

"SEED Labs"

CURATOR'S NOTE — WHY IT EARNS A PLACE

the university-grade lab curriculum: hundreds of hands-on exercises for every security course topic.

WHO USES IT · WHAT IT REPLACES

Used by university courses teaching hands-on security; the free open labs adopted by hundreds of classes.

04 Realm Directory

24 curated entries · A–Z

Android Security Awesome
Bug Bounty Hunter
CTF Field Guide
Cyber Security Base
GTFOBins
hacker-roadmap
Hacker101
Hacksplaining
HackTricks
Infosec Reference
IppSec
LiveOverflow
LOLBAS
Network Segmentation Cheat Sheet
OWASP Cheat Sheet Series
OWASP MSTG
OWASP WSTG
PayloadsAllTheThings
PortSwigger Web Security Academy
Professor Messer
pwn.college
Roppers Academy
SecuriBench Micro
SEED Labs